



Automatic Music Transcription for Guitar: A Machine Learning Approach to Creating a Guitar Practice Tool

Author: Aryan Sunil

Supervisor: Juan Martinez Avila

University of Nottingham

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The Problem

Why existing tools fall short for guitar learners



Inaccurate Tabs Online

User-published tabs are often unverified, leading to frustration for self-taught learners



Curated Libraries Only

Tools like Yousician and SmartMusic only work with their pre-loaded library.



No Ear Training Integration

Existing AMT systems produce tabs but provide no mechanism for users to verify, correct, or learn



Cloud & Privacy Concerns

Commercial tools require uploading audio to servers, raising copyright and privacy issues

Aims & Objectives

Design and evaluate an assistive practice system for guitarists using AI transcription

1. Compare Architectures

Benchmark baseline heuristic vs sequential pipeline (Basic Pitch + 1D CNN) vs end-to-end 2D CNN against a GuitarSet dataset

2. Build Interactive Practice Tool

Develop a GUI with tab editing, MIDI playback comparison, and confidence scoring to support an ear-training workflow

3. Export to Industry Formats

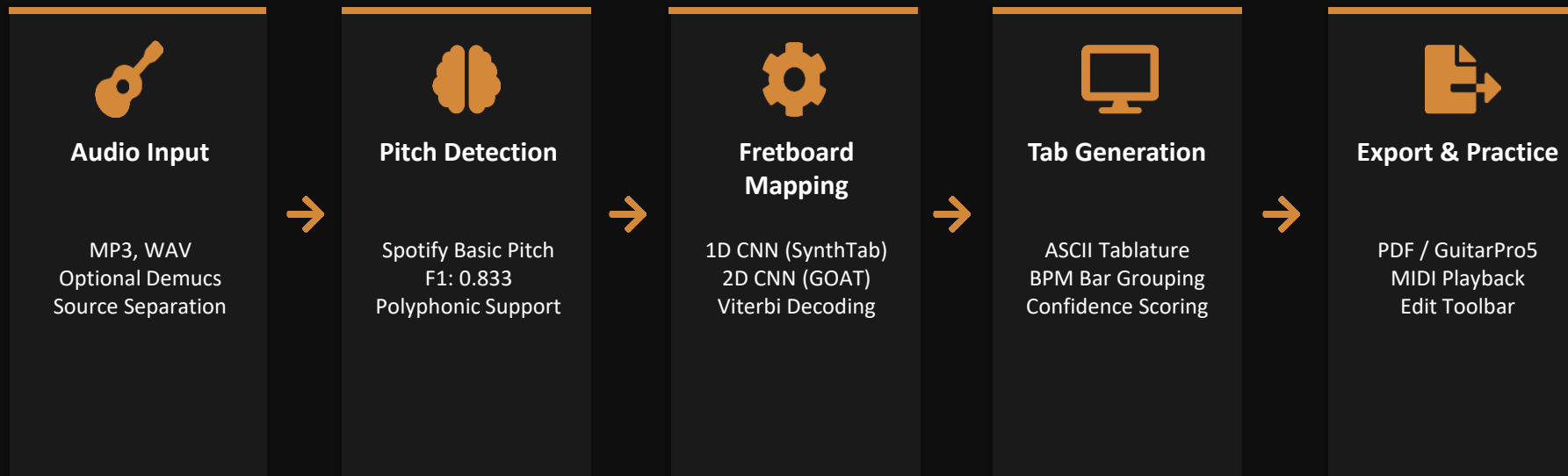
Generate ASCII tablature with PDF and GuitarPro5 export for compatibility with existing guitar tab software

4. Lightweight & Local

Run entirely on consumer hardware with no server-side processing, protecting user copyright and privacy

System Architecture

Three models compared: Heuristic Baseline, Sequential CNN, and End-to-End CNN



Live Demo

Key Results

Evaluated on GuitarSet dataset (60 files) based on exact match: pitch + onset + string + fret

0.833

F1 Pitch Detection
(Basic Pitch + Post-processing)

60.3%

Fretboard Accuracy
(Sequential CNN Pipeline)

+10.7%

Improvement Over
Heuristic Baseline (49.6%)

Model	Accuracy	Avg String Err	Avg Fret Err
Simple Heuristic	49.6%	0.81	3.88
Sequential CNN (Synth)	60.3%	0.43	2.01
End-to-End (GOAT)	17.4%	0.31	1.45

System Workflow

Ear-training workflow: an imperfect transcription becomes an active learning exercise

1

Listen

Upload audio and run transcription through selected model

2

Predict

View generated tablature with traffic light confidence colours guiding attention

3

Compare

Play original audio alongside MIDI playback to identify differences by ear

4

Correct

Edit tabs using the safe toolbar, then export corrected tabs to PDF or GuitarPro5

Contribution & Future Work

Novel contributions and directions for improvement

Contributions

First GMT system targeting interactive ear-training with listen, predict, compare, correct workflow

Comparative analysis: sequential pipeline (60.3%) outperforms end-to-end (17.4%) and heuristic baseline (49.6%)

Traffic light confidence system guides learners to notes most likely needing correction

Lightweight, local-only system protecting user copyright with modular MVC architecture

Future Work

Data augmentation via Spotify Pedalboard (distortion, effects)

Key Limitations

Evaluated only on acoustic guitar (GuitarSet bias)
MIDI playback not tempo-synchronised to original
No formal user evaluation

Thank you for listening!

Q & A